MULTIVARIABLE CALCULUS • UNIT 8.01

C Surface Integrals, Divergence Theorem & Stokes Theorem

PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Surface Integrals, Divergence Theorem & Stokes Theorem Practice Exercises

1 MCQ

Find $\nabla \cdot \mathbf{F}$ at $(2,\ 1,\ 0)$ for $\mathbf{F} = \langle ye^x, xz, yz^2 \rangle$.

A e^2

B $1 + e^2$

C $-e^2$

D $2e^2$

2 MCQ

Find $\nabla \cdot \mathbf{F}$ at $(2,\ -2,\ 1)$ for $\mathbf{F} = \langle xz, yz, xy \rangle$.

A -4

B 2

C 3

D -2

3 MCQ

Find $\nabla \cdot \mathbf{F}$ at $(2,\ -2,\ 0)$ for $\mathbf{F} = \langle ye^x, xz, yz^2 \rangle$.

A $1 - 2e^2$

B $-2e^2$

C $2e^2$

D $-4e^2$

4 MCQ

Find $\nabla \cdot \mathbf{F}$ at $(2,\ 1,\ -2)$ for $\mathbf{F} = \langle xz, yz, xy \rangle$.

A -3

B 4

C -4

D -8

5 MCQ

Find $\nabla \times \mathbf{F}$ at $(2,\ 0,\ 1)$ for $\mathbf{F} = \langle y, -x, z^2 \rangle$.

A $\langle 0, 0, 2 \rangle$

B $\langle 0, -2, 0 \rangle$

C $\langle 0, 0, -2 \rangle$

D $\langle 1, 1, -1 \rangle$

6 MCQ

Find $\nabla \cdot \mathbf{F}$ at $(-2,\ 1,\ -2)$ for $\mathbf{F} = \langle x^2, y^2, z^2 \rangle$.

A 9

B -5

C 6

D -6

7 MCQ

Find $\nabla \times \mathbf{F}$ at $(-2,\ -2,\ 2)$ for $\mathbf{F} = \langle ye^x, xz, yz^2 \rangle$.

A $\langle -6, 0, -2 + e^{-2} \rangle$

B $\langle 6, 0, 2 - e^{-2} \rangle$

C $\langle 0, 2 - e^{-2}, 6 \rangle$

D $\langle 7, 1, 3 - e^{-2} \rangle$

8 MCQ

Find $\nabla \cdot \mathbf{F}$ at $(0,\ -2,\ 0)$ for $\mathbf{F} = \langle y, -x, z^2 \rangle$.

A -2

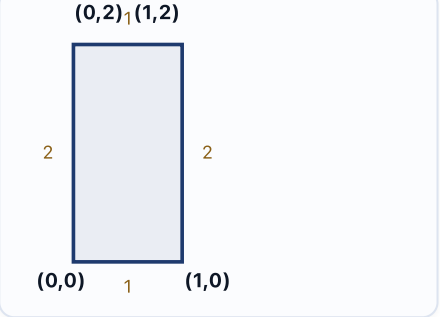
B 0

C 1

D 2

9 MCQ

Consider the surface S which is the portion of the plane $z = 2x + 3y + 1$ lying above the rectangular region $R = [0, 1] \times [0, 2]$ in the xy-plane. Calculate the surface area of S.



A $\sqrt{14}$

B $3\sqrt{14}$

C $2\sqrt{14}$

D $4\sqrt{14}$

10 MCQ

Consider the vector field $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$. Let S be the closed surface of the unit cube defined by $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$, oriented outwards. According to the Divergence Theorem, the flux of $\mathbf{F}$ through S is equal to the triple integral of the divergence of $\mathbf{F}$ over the volume V enclosed by S. Calculate this flux by evaluating $\iiint_V \nabla \cdot \mathbf{F}\, dV$.

A 1

B 2

C 4

D 3

11 MCQ

Find $\nabla \times \mathbf{F}$ at $(0,\ -2,\ -1)$ for $\mathbf{F} = \langle x^2y, -xy^2, z\rangle$.

A $\langle 0, 0, 4\rangle$

B $\langle 0, 0, -4\rangle$

C $\langle 0, -4, 0\rangle$

D $\langle 1, 1, -3\rangle$

12 MCQ

Find $\nabla \times \mathbf{F}$ at $(0,\ -2,\ 2)$ for $\mathbf{F} = \langle xy, yz, xz\rangle$.

A $\langle 2, -2, 0\rangle$

B $\langle -2, 0, 2\rangle$

C $\langle 3, -1, 1\rangle$

D $\langle -2, 2, 0\rangle$

13 MCQ

Calculate $\nabla \cdot \mathbf{F}$ at $(2,\ 2,\ -2)$ for $\mathbf{F} = \langle xz, yz, xy\rangle$.

A -4

B -3

C 4

D -8

14 MCQ

Compute $\nabla \cdot \mathbf{F}$ at $(1,\ 1,\ -2)$ for $\mathbf{F} = \langle xz, yz, xy\rangle$.

A -4

B -3

C 4

D -8

15 MCQ

Find $\nabla \cdot \mathbf{F}$ at $(-2,\ 0,\ 0)$ for $\mathbf{F} = \langle yz, xz, xy\rangle$.

A 1

B 2

C -1

D 0

16 MCQ

Calculate $\nabla \cdot \mathbf{F}$ at $(0,\ -2,\ -1)$ for $\mathbf{F} = \langle ye^x, xz, yz^2\rangle$.

A -4

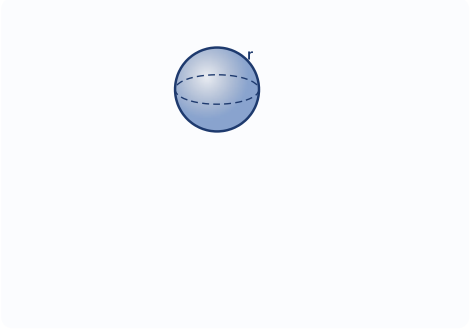
B 2

C 3

D -2

17 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ across the sphere $x^2 + y^2 + z^2 = 9$.



A $\frac{2916\pi}{5}$

B $\frac{5832\pi}{5}$

C $\frac{1458\pi}{5}$

D $-\frac{2916\pi}{5}$

18 MCQ

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle -y, x, 0 \rangle$ around the boundary of the square $[0, 2] \times [0, 2]$ in the plane $z = 0$, oriented counterclockwise seen from above.

A 8

B 16

C -8

D 4

19 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle 2x, -y, 3z \rangle$ across the sphere $x^2 + y^2 + z^2 = 1$.

A $\frac{8\pi}{3}$

B $-\frac{16\pi}{3}$

C $\frac{32\pi}{3}$

D $\frac{16\pi}{3}$

20 MCQ

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle -y, x, z \rangle$ around the boundary of the square $[0, 1] \times [0, 1]$ in the plane $z = 0$, oriented counterclockwise seen from above.

A -2

B 1

C 4

D 2

21 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x + y, y + z, x + z \rangle$ across the sphere $x^2 + y^2 + z^2 = 9$.

A -108π

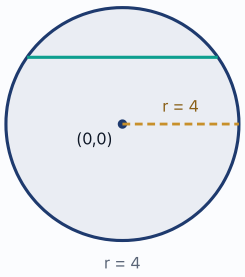
B 216π

C 54π

D 108π

22 MCQ

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle -2y, 2x, y \rangle$ around the circle $x^2 + y^2 = 1$ in the plane $z = 0$, oriented counterclockwise seen from above.



A 8π

B 4π

C -4π

D 2π

23 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x, y, z \rangle$ across the sphere $x^2 + y^2 + z^2 = 4$.

A 64π

B 32π

C 16π

D -32π

24 MCQ

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x, y, z \rangle$ across the closed cylinder $x^2 + y^2 \leq 1,\ 0 \leq z \leq 1$.

A $\frac{3\pi}{2}$

B -3π

C 6π

D 3π

25 MCQ

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle -3y, x, z \rangle$ around the circle $x^2 + y^2 = 1$ in the plane $z = 0$, oriented counterclockwise seen from above.

A 2π

B 8π

C 4π

D -4π